

Josh Yeh

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Career Objective

Electrical Engineering and Engineering Management student with hands-on experience in PCB assembly, electronics testing, semiconductor fabrication, controls, and electromechanical systems.

Education

Washington University | St. Louis, MO

B.S. in Electrical Engineering – GPA: 3.4/4.0

M.Eng. in Engineering Management

Pepperdine University | Malibu, CA

B.S. in Physics; B.A. in Natural Science

August 2024 – Present

Expected Graduation May 2027

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January 2021 – April 2024

Skills

Programming & Analysis: Python, MATLAB, R

Hardware & PCB: EasyEDA, PCB Design & Layout, PCB Assembly/Rework, Soldering, PSpice, Multisim

Lab & Test Equipment: Oscilloscopes, Multimeters, Power Supplies, Function Generators

Mechanical & Controls: SolidWorks, Embedded Systems, PID Control, Stepper Motors, Motion Control

AI & Data: PyTorch, Label Studio, Hugging Face, VLM Benchmarking

Projects

Two-Axis Cryogenic Goniometer for Graphene Experiment (Capstone)

January 2025 – May 2026

- Led systems integration for a goniometer enabling precise rotation in cryogenic, high magnetic field environments
- Developed and tested motor-control electronics using stepper drivers, custom PCBs, and SCPI-based controls
- Analyzed and mitigated stepper motor error sources, including dead reckoning drift, step loss, and backlash

Semiconductor Fabrication Lab – Silicon Photodiode Fabrication

January 2025 – May 2026

- Fabricated silicon p-n photodiodes in a Class 100/1000 cleanroom using full process flow
- Maintained industry-standard wafer runsheets and digital lab notebooks, documenting process parameters
- Characterized fabricated devices using electrical test equipment and analyzed measurement results

Meta Glasses AI Assistant (Python)

May 2025 – June 2025

- Engineered real time assistive eyewear on Meta's Aria glasses, delivering concise scene descriptions in <1 s
- Integrated Whisper (STT), Qwen (VLM), and TTS into a multimodal pipeline through a Flask server
- Reduced captioning latency by 40% through GPU inference optimization and efficient image resizing

Autonomous Robot (Python)

September 2024 – December 2024

- Designed control system for Raspberry Pi car with integrated sensors and actuators for autonomous driving
- Coded vision-based navigation using camera to detect obstacles and make real-time directional adjustments
- Developed proportional integral derivative controller to maintain target speed by adjusting motor PWM signals

Experience

Electrical Engineering Researcher | Henriksen Lab, Washington University in St. Louis

May 2026 – Present

- Continuing capstone development of a motorized two-axis goniometer for graphene experiments
- Refining motion control software and user interface for angular positioning, calibration and safety constraints
- Evaluating cryogenic compatibility, mechanical stability, thermal load, and wiring strain for real world integration

AI Research Assistant | Harvard Ophthalmology AI Lab

2025 – 2026

- Built "VISTA," a vision-audio dataset for VLM benchmarking and fine-tuning using Meta Aria smart glasses
- Managed and trained a team of annotators, creating guidelines for high-quality annotations in Label Studio
- Designed annotation tools and benchmarking scripts, hosted data via Hugging Face and GitHub

Electrical Engineering Grader | Washington University in St. Louis

September 2025 – December 2025

- Assisted course staff by grading assignments and exams for *Introduction to Electrical and Electronic Circuits*
- Applied detailed rubrics to evaluate circuit analysis, design, and problem-solving accuracy
- Strengthened technical expertise by reviewing a wide range of analog and digital circuit applications

Design & Construction Jr. Project Assistant | Pepperdine University

2021 – 2024

- Conducted audits of active construction projects to ensure compliance and progress tracking
- Organized and streamlined file management systems, improving accessibility and efficiency
- Coordinated with vendors, project stakeholders, and university management to support project execution and resolve operational needs.